Alphabet Investment Thesis

Alphabet Inc. | Share class [GOOGL / GOOG]

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Valuation date 19/10/2026

Base Case Metric	Model Result
Estimated Value per share (DCF Value)	$93.24
Market price at valuation date	$320.21
Implied upside or downside	70.9%
WACC	9.82%
Terminal growth rate	2.5%

1 Executive summary

This research assesses whether growth in Google Cloud and AI monetisation can generate sufficient cash flow to support Alphabet's valuation while sustaining profitability in its Search business.

My investment thesis centres on three propositions: Search can remain a durable source of cash generation; Google Cloud can contribute a greater share of operating profit; and the value created by AI will depend on whether incremental cash flows justify the associated infrastructure investment.

The analysis tests these propositions through historical financial analysis, five-year forecasts, discounted cash flow valuation and comparable company analysis. An investment recommendation will follow from the completed valuation, scenario analysis and assessment of the principal risks.

2 Company overview

Business Model
Alphabet, Google's parent company, specialises in digital advertising, online services, cloud computing and artificial intelligence. It operates through three segments: Google Services, which includes Search, YouTube and Android; Google Cloud, which provides business computing and AI services; and Other Bets, which includes emerging ventures such as Waymo. Google Services generates most of its revenue and profit, while Cloud is its fastest-growing segment. Its future performance depends on maintaining its advertising strength and converting investment in AI and cloud infrastructure into sustainable cash flow.

In FY2025 alone, Revenues were $402.8 billion, an increase of 15% year over year, primarily driven by an increase in Google Services revenues of $37.8 billion, or 12%, and an increase in Google Cloud revenues of $15.5 billion, or 36%.

Financial metric	Latest completed financial year
Revenue	$402.836
Revenue growth	15%
Operating income	$129.04bn
Operating margin	32%
Net Income	$137,031

3 Investment case summary

Thesis pillar	Proposition to test	Forecast connection
Search resilience	Preserve advertising demand and monetisation as search behaviour evolves.	Consolidated growth and operating margin.
Cloud profitability	Grow Cloud while improving or sustaining operating economics.	Growth, gross margin and operating expenses.
AI investment returns	Additional revenue and efficiencies outweigh investment and operating costs.	CapEx, depreciation, margins and terminal FCFF.

4 Investment thesis

4.1 Search resilience is central to the investment case

My first proposition is that Search can retain sufficient advertising demand as the way users obtain information changes. Greater AI usage only supports the investment case if it sustains advertising economics or produces another profitable revenue stream. Higher query volumes would add less value if computing cost rises faster than monetisation.

The base forecast requires consolidated growth of 14% in FY2026, easing to 10% in FY2030. Search disruption would challenge that path and could reduce the operating margin below its projected 31.9% to 34.1% range. Sustained weak monetisation or rising delivery costs would therefore undermine both growth and cash conversion.

4.2 Google Cloud could become a more significant profit contributor

My second proposition is that expanding Cloud demand can support the group's profitability. Customer acquisition, pricing and utilisation matter alongside revenue. Faster sales would add limited value if discounts, capacity commitments or operating expenses absorb the incremental profit.

The model expresses this proposition indirectly through gross margin rising from 59.5% to 60.4%, while R&D and sales spending decline relative to revenue. This should be tested against subsequent segment disclosures. The model does not demonstrate that Cloud alone delivers the assumed margin improvement.

4.3 AI creates value only if the investment earns sufficient returns

My third proposition is that AI monetisation and efficiencies can justify the associated infrastructure investment. In the model, CapEx falls from 30% to 20% of revenue while depreciation rises from 6.0% to 7.4%. This combination drives a marked recovery in free cash flow after an initially investment-heavy period.

The central risk is timing: revenue and margins may improve before cash returns cover the investment. I would strengthen this proposition only when greater cash generation accompanies spending, rather than infer success from expenditure or product adoption alone. AI benefits are embedded in the consolidated assumptions, with no additional AI premium added to enterprise value.

5 Valuation

I use workbook's five-year FCFF forecast for FY2026–FY2030 and perpetual-growth terminal value. Historical coverage is FY2023–FY2025. Figures below reproduce its operating assumptions; they are estimates to be justified, not management guidance.

Driver	2026E
Revenue growth	14.0%
Gross margin	59.5%
R&D / revenue	15.2%
Sales / revenue	7.1%
G&A / revenue	5.3%
Operating margin	31.9%
Tax rate	18.0%
CapEx / revenue	30.0%
D&A / revenue	6.0%

Revenue increases from $402.84bn in FY2025 to $709.65bn in FY2030, equivalent to approximately 12.0% annual compound growth. Operating income reaches $241.99bn. The improvement requires both continuing demand and operating cost discipline; it is not simply an extrapolation of unchanged profitability.

5.3 Discount rate and terminal value

The workbook calculates a 9.821% WACC using a 4.61% risk-free rate, beta of 1.10, 4.80% equity risk premium and 4.90% pre-tax cost of debt. With an 18% tax rate, after-tax debt cost is 4.018%. Its capital weights are approximately 98.83% equity and 1.17% debt. These are model inputs and proxies, not independently verified market estimates for 26 October 2026.

Terminal FCFF is $103.41bn, calculated as FY2030 FCFF × 1.025. Terminal enterprise value is terminal FCFF divided by WACC less 2.5% growth. The 2.5% rate remains below WACC, but sustainable long-term reinvestment must still be assessed. Capital intensity and profitability inherited from FY2030 materially affect this estimate.

5.4 Enterprise to equity value

Valuation bridge	USD bn unless stated
PV of forecast FCFF	175.83
PV of terminal enterprise value	884.23
Enterprise value	1,060.05
Add cash and short-term investments	126.84
Less debt included by the model	46.55
Equity value	1,140.35
Diluted shares in millions	12,230
Implied value per share	$93.24
Entered share price	$320.21
Implied return	−70.9%

6 Scenarios and sensitivity

Case	Value/share	Return vs $320.21	2030 revenue USD bn	2030 FCFF USD bn
Bear	$56.36	-82.4%	619.55	61.32
Base	$93.24	-70.9%	709.65	100.89
Bull	$138.55	-56.7%	809.94	150.26

Bear case

Weaker Search monetisation or Cloud demand slows consolidated growth, while cost pressure reduces margins and infrastructure spending remains high. The resulting $56.36 value implies 82.4% downside. This case tests an unfavourable combination of weaker growth and poor capital efficiency rather than a single isolated shock.

Base case

The supplied forecast produces $93.24 per share and 70.9% downside. Growth moderates gradually, margins improve and CapEx falls as a share of revenue. It therefore already incorporates a meaningful improvement in operating and investment efficiency.

Bull case

Stronger monetisation, operating leverage and lower capital intensity produce $138.55 per share and 56.7% downside. The entered price remains above this tested outcome. This does not bound every possible upside case: the stresses are analyst-selected, while excluded assets and valuation-date corrections could also affect value.

7 Catalysts and indicators to monitor

Potential catalysts must represent developments that could change expectations, rather than simply describe desirable outcomes.

Potential catalyst	Evidence and forecast implication

Improved AI monetisation	Stronger monetisation accompanied by operating profit would support revenue and margin assumptions. Usage alone is insufficient.
Cloud profit conversion	Segment growth with better incremental profitability would support consolidated operating leverage; growth bought through discounts would weaken it.
Higher capital efficiency	CapEx growing slower than revenue with stable capacity and service quality would support the forecast decline in CapEx/revenue.
Regulatory clarity	A disclosed decision with measurable operational consequences would allow specific changes to revenue, costs, cash claims or distribution assumptions.

8 Principal risks

8.1 Search disruption

Competing AI assistants or other platforms could reduce query share, advertising engagement or revenue per query. Even if user activity remains strong, more expensive delivery could weaken profitability. This risk affects consolidated revenue growth, gross margin and operating expenses. A structural loss of competitive strength could also weaken sustainable terminal cash flow. I would test lower growth and margins together where the same disruption affects both.

8.2 Infrastructure investment returns

Infrastructure could become available before demand, remain underutilised or need replacement earlier than expected. This challenges the assumed decline in CapEx from 30% to 20% of revenue and the forecast margin expansion. Slower monetisation should delay forecast revenue benefits; persistent spending should reduce FCFF. Depreciation reduces EBIT but is added back in FCFF, so it must not be treated as a second cash investment outflow. Asset lives, tax treatment and replacement spending require consistent assumptions.

8.3 Cloud competition

Discounting, customer acquisition costs or weaker demand could limit both growth and profitability. Committed capacity may prevent expenditure from falling immediately when sales slow. In this consolidated model, the affected assumptions are revenue growth, gross margin, R&D and selling expense ratios, and CapEx. A future segment build would allow a more precise test of Cloud's contribution. The bear case captures the direction of these effects without claiming a specific Cloud valuation.

8.4 Model uncertainty and omitted value

Errors in historical taxes, cash flows, equity signs and schedules limit confidence in the model. Narrow working capital, fixed shares, incomplete nonoperating asset adjustments and mismatched valuation timing can also distort equity value. The direction of the net correction is uncertain. Higher WACC or lower terminal growth reduces value, but lower cash-flow forecasts should not automatically be paired with a higher WACC for the same risk without justification.

The 70.9% downside is conditional on the supplied assumptions and bridge. It should not be presented as a precise expected price fall or investment outcome. Appendix A identifies concrete reconciliation work needed before the model can support a firm recommendation.

9 Thesis reassessment

I would become more positive if evidence supported stronger sustainable cash generation than assumed in the base case, or if the share price offered a greater margin of safety. I would become more cautious if Search economics weakened, Cloud growth failed to translate into profit, or investment requirements increased without sufficient additional cash flow.

A change in share price alone would not establish whether the operating thesis was correct; I would compare business performance with the original forecasts.

10 Conclusion

The supplied DCF indicates $93.24 per share compared with its entered market price of $320.21, or 70.9% downside. On those inputs, the model-implied recommendation is SELL. The operating stresses produce $56.36 in the bear case and $138.55 in the bull case; the WACC/growth grid ranges from $74.19 to $126.61.

This research is prepared for educational purposes and does not constitute investment advice. Forecasts are estimates, and the working draft does not yet establish a recommendation.